

Chapter 6 - Testing and Results

6.1 Introduction

Testing is a critical phase of the software development lifecycle that verifies whether the developed application satisfies the specified functional and business requirements. The AI-Powered Customer Support & Service Automation system was tested to ensure that each Salesforce component operated correctly and that all business processes executed as expected. Testing included verification of custom objects, field validations, automation flows, queues, Prompt Builder functionality, Lightning App navigation, and overall system integration. The objective was to ensure reliable performance, accurate data processing, and smooth user interaction.

6.2 Testing Strategy

The application was tested using sample customer, product, order, and support case records. Individual modules were tested independently before conducting end-to-end system testing.

The following testing methods were performed:

- Functional Testing
- Validation Rule Testing
- Flow Testing
- Queue Assignment Testing
- Prompt Builder Testing
- User Interface Testing
- Integration Testing

6.3 Test Cases

Test ID	Test Scenario	Expected Result	Status
TC-01	Create Customer Record	Customer record created successfully	Pass
TC-02	Create Product Record	Product record created successfully	Pass
TC-03	Create Order Record	Order linked to Customer and Product	Pass
TC-04	Create Use Case	Record created successfully	Pass
TC-05	Validation Rule for Damage	Error displayed when Priority is missing	Pass
TC-06	Validation Rule for Warranty	Error displayed when Priority is missing	Pass
TC-07	High Priority Flow	Record assigned to Damage Support Queue	Pass
TC-08	Prompt Builder	AI Summary generated successfully	Pass
TC-09	Lightning Application	Navigation works correctly	Pass
TC-10	Queue Assignment	Queue ownership updated correctly	Pass

6.4 Validation Rule Testing

Validation rules were tested by intentionally entering incomplete information.

Scenario 1

- Issue Category = Damage
- Priority = Blank

Result: The system displayed the validation error: "Priority is mandatory for Damage cases."

Scenario 2

- Issue Category = Warranty
- Priority = Blank

Result: The system prevented record creation and displayed the appropriate validation message.

6.5 Flow Testing

The Record-Triggered Flows were tested by creating Use Case records with different priorities.

Test Scenario

- Priority = High

Result: The flow executed successfully and assigned the record to the Damage Support Queue.

6.6 Prompt Builder Testing

Prompt Builder was tested using sample customer support records.

Input

- Product
- Order
- Issue Category
- Priority

Output

A professional AI-generated summary was successfully stored in the AI Summary field.

6.7 User Interface Testing

The Lightning Application was tested to verify:

- Navigation
- Object Accessibility
- Record Creation
- Dynamic Forms
- AI Summary Display

All user interface components functioned correctly.

6.8 Expected vs Actual Results

Module	Expected Result	Actual Result
Customer	Record Created	Successful
Product	Record Created	Successful
Order	Lookup Working	Successful
Use Case	Record Saved	Successful
Validation Rules	Error Generated	Successful
Flows	Auto Assignment	Successful
Queues	Owner Updated	Successful
Prompt Builder	AI Summary Generated	Successful
Lightning App	Navigation Working	Successful

6.9 Results

The application successfully achieved the project objectives.

Major outcomes include:

- Customer support automation implemented.
- Validation rules prevented incorrect data entry.
- Record-Triggered Flows automated business processes.
- Prompt Builder generated AI-assisted summaries.
- Queue assignment reduced manual intervention.
- Lightning Application improved usability.
- Overall system performance met business expectations.

6.10 Chapter Summary

Testing confirmed that all major Salesforce components operated correctly and integrated successfully. The AI-powered customer support application demonstrated reliable automation, accurate validation, effective AI assistance, and secure data management.